

Francesco Arzani

Researcher in quantum information and quantum optics



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Academic and professional experience

- 09/2023–current **Junior professor**, INRIA Paris and École Normale Supérieure – Paris, France, QAT team
- 05/2021–09/2025 **Consultant**, Xanadu Quantum Technologies – Toronto, Ontario, Canada (remote)
- 09/2020–08/2023 **Post-doctoral researcher**, Freie Universität Berlin – Quantum information theory, quantum many-body theory, and quantum optics group,
Supervisor: Prof. Jens Eisert
- 10/2018–10/2019 **Post-doctoral researcher**, MOCQUA team – Loria, Nancy & Quantum Information group – Laboratory of Informatics of Paris 6,
Supervisors: Dr. Damian Markham, Dr. Simon Perdrix
- 04/2018–9/2018 **Post-doctoral researcher**, Université Paris Diderot & Paris center for quantum computing,
Supervisors: Dr. Damian Markham, Dr. Iordanis Kerenidis

Education

- 01/2015–03/2018 **PhD in Physics**, École Normale Supérieure, Paris; Kastler Brossel Laboratory,
Thesis: “Measurement-based quantum information with optical frequency combs”,
Supervisors: Prof. Nicolas Treps & Prof. Claude Fabre
Funding: Paris Sciences Lettres Research University
- 10/2012–12/2014 **Master’s degree in Theoretical Physics**, Università degli Studi di Pavia,
Master’s thesis: “Measurement-based quantum information with optical frequency combs”,
Supervisor: Prof. Chiara Macchiavello; Co-supervisors Prof. Nicolas Treps & Dr. Giulia Ferrini
Grade: 110/110 cum laude (with honours)
- 02/2014–08/2014 **Master’s internship**, Quantum Optics group – Kastler Brossel Laboratory, Université Paris 6 Pierre et Marie Curie
- 09/2013–07/2014 **ERASMUS exchange period**, Université Paris 6 Pierre et Marie Curie
- 09/2009–10/2012 **Bachelor’s Degree in Physics**, Università degli Studi di Pavia,
Bachelor’s thesis: “An informational approach to quantum thermodynamics beyond the von Neumann Regime”,
Supervisor: Dr. Paolo Perinotti
Grade: 110/110
- 2009 **High school diploma**, Liceo scientifico tecnologico, IIS “Guglielmo Marconi”, Tortona
Grade: 100/100 cum Laude (with honours)

Grants, awards, fellowships

- 03/2025–02/2028 **ANR-PRCI**, Franco-German collaborative research grant on Bosonic Lattice Codes (BoLaCo),
French coordinator, partnered with Freie Universität Berlin
- 09/2020–08/2022 **Humboldt postdoctoral fellowship**, Freie Universität Berlin
- 09/2009–07/2015 **Student at IUSS**, Istituto Universitario di Studi Superiori, Pavia
- 09/2009–07/2014 **Student at Collegio Ghislieri**, Pavia, Residence for university students with merit based admission

Teaching and outreach

- 10/2024– **Instructor**, École Normale Supérieure – Paris Sciences Lettres ,
Quantum computing (M2), Quantum engineering master

- 10/2023–02/2024 **Instructor**, *École Normale Supérieure – Paris Sciences Lettres*, Quantum programming (M2), Quantum engineering master
- 11/2021–03/2022 **Teaching assistant**, *Freie Universitat Berlin*, Quantum information (Master’s level)
- 11/2019 **Fête de la Science**, *Sorbonne Université*, Presentation of quantum information to high school students
- 09/2018–12/2018 **Teaching assistant and lab demonstrator (Vacataire)**, *Université d’Évry val d’Essonne*, Introduction to experimental physics (Bachelor’s level)

Academic reviewing

Phys. Rev. Lett., Phys. Rev. A, Quantum Journal, JOSA B, Eur. Phys. J. D, Optics Express, QIP (conference), QCTiP (conference)

Languages

- Native Italian
- Fluent English, French
- Basic German

Computer Skills

- Operating systems Unix/Linux, Windows
- Programming languages Mathematica, Python, Julia, C++

Publications

Preprints

- *Quantum low-density lattice codes*
T. Hillmann, J. Eisert, **F. Arzani** (arxiv:2609.03021).
- *Improved GKP magic states from error-corrected non-Gaussian quantum states*
S. David, J. Davis, U. Chabaud, **F. Arzani** (arxiv:2607.07833).
- *A hidden bottleneck in classical and quantum linear reservoir computing*
J. Nokkala, F. Centrone, **F. Arzani** (arxiv:2605.29071).
- *On the complex zeros of the wavefunction*
S. Cerf, C. Wassner, J. Davis, **F. Arzani**, U. Chabaud (arxiv:2507.23468).
- *Detecting quantum non-Gaussianity with a single quadrature*
C. Wassner, J. Davis, S. Cerf, U. Chabaud, **F. Arzani** (arxiv:2507.23005).
- *Towards fault-tolerant quantum computation with universal continuous-variable gates*
S. Blair, **F. Arzani**, G. Ferrini, A. Ferraro (arxiv:2506.13643).

Journal articles (peer reviewed)

- *Effective descriptions of bosonic systems can be considered complete*
F. Arzani, R. I. Booth, U. Chabaud
Nature Communications 16, 9744 (2025). (arxiv:2501.13857).
- *Exploiting symmetry in variational quantum machine learning.*
J. J. Meyer, M. Mularski, E. Gil-Fuster, A. A. Mele, **F. Arzani**, A. Wilms, J. Eisert
Physical Review X, 010328 (2023). (arxiv:2205.06217).
- *Gottesman-Kitaev-Preskill codes: A lattice perspective.*
J. Conrad, J. Eisert, **F. Arzani**
Quantum 6, 648 (2022). (arxiv:2109.14645)
- *Random coding for sharing bosonic quantum secrets.*
F. Arzani, G. Ferrini, F. Grosshans, D. Markham
Physical Review A 100 (2), 022303 (2019). (arXiv:1808.06870)
- *Bloch-Messiah reduction for twin beams of light.*
D. B. Horoshko, L. La Volpe, **F. Arzani**, N. Treps, C. Fabre, M. I. Kolobov
Physical Review A 100 (1), 013837(2019). (arXiv:1903.06578)
- *High-dimensional quantum encoding via photon-subtracted squeezed states.*
F. Arzani, A. Ferraro, V. Parigi
Physical Review A 99 (2), 022342 (2019). (arXiv:1811.09263)
- *Violating Bell inequalities with entangled optical frequency combs and multi-pixel homodyne detection.*
W. N. Plick, **F. Arzani**, N. Treps, E. Diamanti, D. Markham
Physical Review A 98 (6), 062101 (2018). (arXiv:1805.06059)
- *Reconfigurable optical implementation of quantum complex networks.*
J. Nokkala, **F. Arzani**, F. Galve, R. Zambrini, S. Maniscalco, J. Piilo, N. Treps, V. Parigi.
New Journal of Physics 20 (5), 053024 (2018). (arXiv:1708.08726)
- *Versatile engineering of multimode squeezed states by optimizing the pump spectral profile in spontaneous parametric down-conversion.*
F. Arzani, C. Fabre, N. Treps.
Physical Review A 97 (3), 033808 (2018). (arXiv:1709.10055)
- *Polynomial approximation of non-Gaussian unitaries by counting one photon at a time.*
F. Arzani, N. Treps, G. Ferrini
Physical Review A 95 (5), 052352 (2017). (arXiv:1703.06693)
- *Multimode entanglement in reconfigurable graph states using optical frequency combs.*
Y. Cai, J. Roslund, G. Ferrini, **F. Arzani**, X. Xu, C. Fabre, N. Treps.
Nature Communications 8, 15645 (2017). (arXiv:1605.02303)
- *A direct approach to Gaussian measurement based quantum computation.*

G. Ferrini, J. Roslund, **F. Arzani**, C. Fabre, N. Treps.
Physical Review A 94 (6), 062332 (2016). (arXiv:1605.03350)

- *Optimization of networks for measurement-based quantum computation.*
G. Ferrini, J. Roslund, **F. Arzani**, Y. Cai, C. Fabre, N. Treps.
Physical Review A 91 (3), 032314 (2015). (arXiv:1407.5318)

Other

- *Harmonizing continuous noise to build a modular photonic quantum computer.*
F. Arzani
Quantum Views 5, 51 (2021).
- *Measurement based quantum information with optical frequency combs.*
F. Arzani
PhD Thesis, Paris Sciences Lettres and École Normale Supérieure de Paris (2018)

Talks, conferences, workshops, schools

- 03/2026 **PCQT**,
Invited talk *Paris center for quantum computing workshop, Paris, France* ,
“Continuous-Variable Quantum Computing and Error Correction”
- 11/2025 **BBQ**,
Invited talk *Byron Bay quantum workshop, New South Wales, Australia* ,
“GKP codes with generalized concatenation or none at all”
- 05/2025 **QUEST Lab**,
Guest seminar *(online) North Carolina State University, Raleigh (NC), USA* ,
“Can effective descriptions of bosonic systems be considered complete?”
- 04/2025 **QCTIP**,
Poster *Conference on quantum computing theory in practice, Freie Universität Berlin, Berlin, Germany*,
“Can effective descriptions of bosonic systems be considered complete?”
- 10/2024 **INRIA and ENS Paris-Saclay**,
Contributed talk *Workshop of the Défi Equip project, ENS Saclay, Saclay, France*,
“Bosonic quantum error correction meets lattice theory”
- 07/2024 **IFISC**,
Tutorial talk *Mini workshop on Non-Gaussian Quantum Reservoir Computing, Palma de Mallorca, Spain*,
“Gaussian and non-Gaussian resources in quantum optics”
- 02/2024 **MC2**,
Guest seminar *Wallenberg Centre for Quantum Technologies, Gothenburg, Sweden* ,
“Springs and crystals: Bosonic quantum error correction meets lattice theory”
- 09/2022 **CQIQC IX**,
Contributed talk *Conference on quantum information and quantum control, Fields Institute, Toronto, Canada*,
“Gottesman-Kitaev-Preskill bosonic error correcting codes: a lattice perspective”
- 09/2021 Guest **Xanadu**,
seminar *(online) Xanadu Quantum Technologies, Toronto ON, Canada*,
“Gottesman-Kitaev-Preskill bosonic error correcting codes: a lattice perspective”
- 09/2021 **IPP**,
Invited talk *Télécom Paris - Institut Polytechnique de Paris, Palaiseau, France*,
“Continuous-variable quantum information, multi-mode quantum optics and bosonic error correcting codes ”
- 09/2021 **SAMOP21**,
Contributed talk *Virtual DPG-Meeting of the Atomic, Molecular, Plasma Physics and Quantum Optics Section (SAMOP)*,
“Gottesman-Kitaev-Preskill bosonic error correcting codes: a lattice perspective”
- 08/2019 **CQIQC VIII**,
Contributed talk *Conference on quantum information and quantum control, Fields Institute, Toronto, Canada*,
“Random coding for sharing bosonic quantum secrets”
- 11/2018 **Q-Turn 2018**,
Contributed talk *Workshop on changing paradigms in quantum science, Universidade Federal de Santa Catarina, Florianópolis, Brazil*,
“Quantum secret sharing using squeezing and almost any passive interferometer”
- 11/2018 **JIQ 2018**,
Contributed talk *Journées d’Informatique Quantique, Laboratoire Lorrain de recherche en informatique et ses applications, Nancy, France*,
“Quantum secret sharing using squeezing and almost any passive interferometer”
- 09/2018 **QuTech 2018**,
Contributed talk *Quantum technology international conference, 1st edition, Kastler Brossel laboratory, Paris, France*,
“Versatile engineering of multimode squeezed states by optimizing the pump spectral profile in spontaneous parametric down-conversion”

- 09/2017 **Workshop on quantum science and technologies**,
 Contributed talk *International Center for Theoretical Physics, Trieste, Italy*,
 “Polynomial approximation of non-Gaussian unitaries by counting one photon at a time”
- 05/2017 **Photons beyond qubits 2017**,
 Invited talk *Palacký University, Olomouc, Czech Republic*,
 “Measurement-based quantum information protocols with optical frequency combs”
- 04/2017 **Quantum information and measurement 2017**,
 Contributed talk *4th edition, Université Pierre et Marie Curie, Paris, France*,
 “Shaping the Pump of a Synchronously Pumped Optical Parametric Oscillator for Continuous - Variable Quantum Information”
- 11/2016 **GDR Quantum Information 2016**,
 Poster *7th colloquium of the CNRS research network on Quantum information, foundations and applications, Télécom ParisTech, Paris, France*,
 “Shaping the Pump of a Synchronously Pumped Optical Parametric Oscillator for Continuous - Variable Quantum Information”
- 06/2016 **CEWQO 2016**,
 Poster *23rd central european workshop on quantum optics, Orthodox Academy of Crete, in Kolymbari, Crete, Greece*,
 “Quantum Computing with Optical Frequency Combs”
- 04/2015 **International School on Parametric Nonlinear Optics**,
 Attendance *Les Houches School of Physics, Les Houches, France*